Principal component analysis

This is an initial example with Allison's art about capturing most of the data if you were a whale shark going for that cluster of Krill.

- If you were a whale shark, what direction would you move to capture the most Krill in one pass?
- What would be the second direction after that one?

With many variables, sometimes it can be useful to simplify our view of our coordinate system to capture *as much about the data as possible* *in as few dimensions as possible*.

In PCA, the axes (principal components) are chosen based on the direction of the data with greatest variance, therefore explaining the most variance using a simplified number of dimensions.

PCA: new axes are created (linear combinations of the original variables) so that the first principal component (PC1) is the direction accounting for the most variance in the multivariate data, the second principal component (PC2) accounts for the second direction with most variance, and so on.

If PC1, PC2 explain most of the variance, we'd still be seeing most of the important things about our data if we just use PC1 and PC2 to analyze our data..

Same number of principal components and as variables in our data.

Example: - 3 dimensional data in an ellipsoid shape

- show 3 axes with PC1, PC2, PC3
- show how data looks like when projected onto PC1, PC2

This is <u>dimensionality reduction</u>: convert complex multidimensional data into fewer dimensions to capture as much information about the data as simply as possible.

Principal component regression approach.

<u>Guiding idea</u>: often most of the variability in the predictors is captured along a few directions, and these directions also capture most of the relationship of the predictors with the response.

<u>In other words</u>: we assume that the directions in which $X_1, \dots, X_p$ show the most variation are the directions that are associated to Y.

This is a reasonable enough approximation to give good results.

In this section we talk about how PCA is used as a dimensionality reduction approach. We need to use the previous dataset from lasso and ridge. and include examples of the intermediate steps in this approach. The final discussion is brief about using CV to find an optimal number of components for the regression.

complete dataset → find principal components → project dataset on principal components to obtain reduced dimension dataset → fit the new model using this new data and the response variable